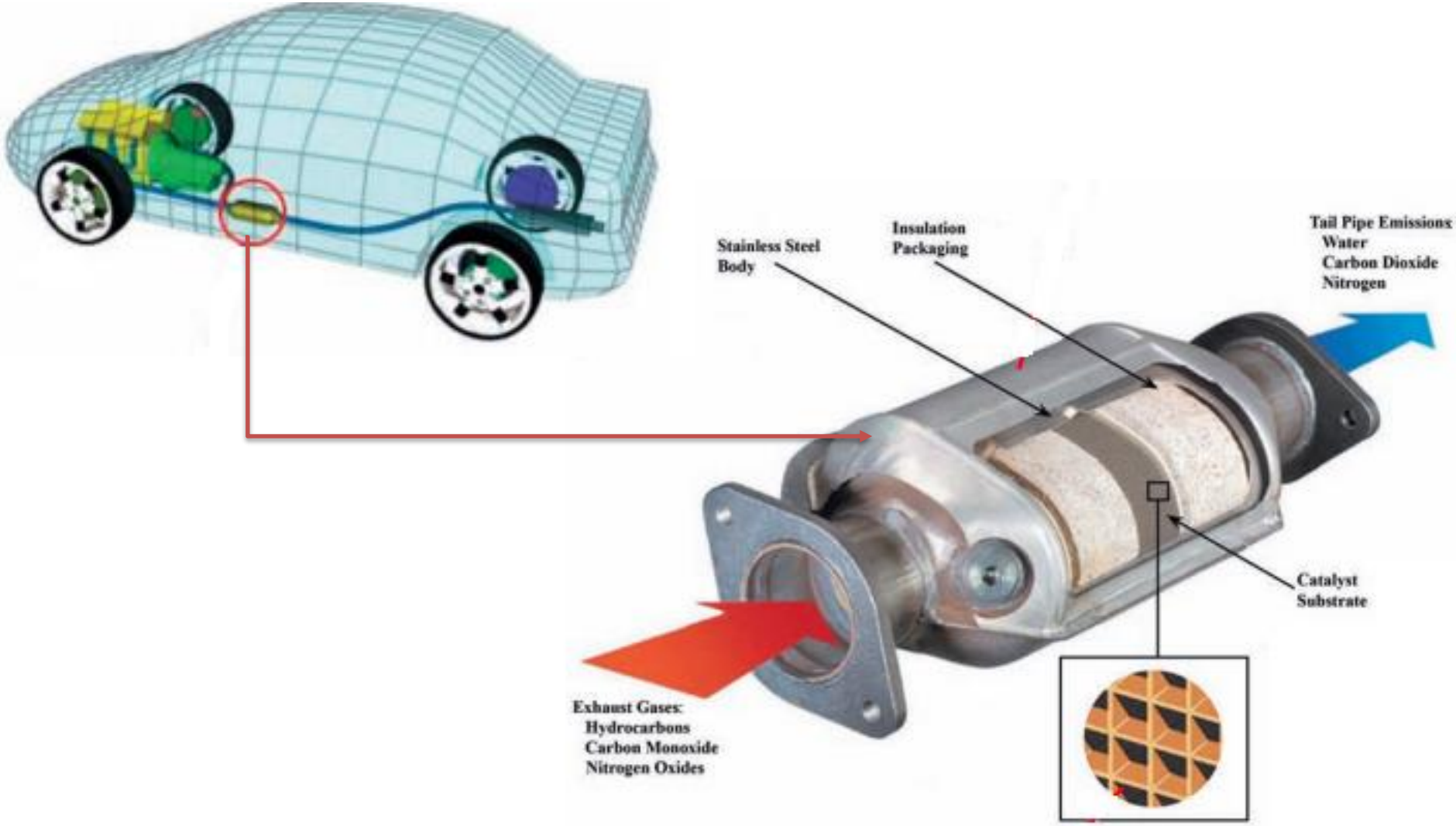
Kwame Nkrumah University of
Science & Technology, Kumasi, Ghana

ME 281 – Engineering Materials

Lecture 4: Imperfections In Solids

S. Boahen (PhD)

Why Study Imperfections In Solids



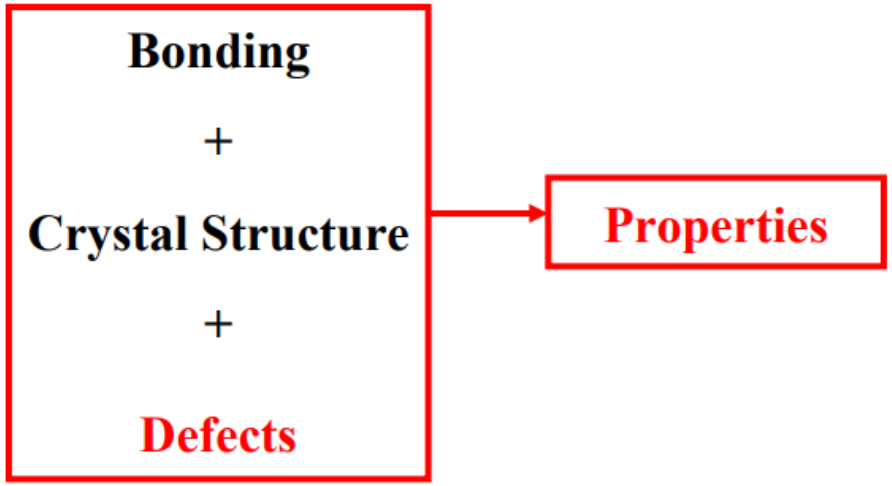
Why Study Imperfections In Solids

- The **properties of some materials** are profoundly influenced by the presence of imperfections.
- For the processing of silicon as a semiconducting material, it is important to specify impurity concentration in appropriate units.
- Development of the desirable mechanical properties for steel alloys relies on the presence of specific impurities, some of which form solid solutions.

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Why are Defects Important

- Defects have a profound impact on the various properties of materials.
- Production of advanced semiconductor devices involve introduction of specific defects in small areas of the sample.
- Defects are responsible for colour (& price) of a diamond crystal.
- Forging a metal tool introduces defects and increases strength of the tool.



Lesson Objectives

- Explain the various imperfections in solid crystals
- Describe both vacancy and self-interstitial crystalline defects.
- Calculate the equilibrium number of vacancies in a material at some specified temperature, given the relevant constants.
- Explain edge, screw, and mixed dislocations
- Understand the atomic structure within the vicinity of
  - ✓ a grain boundary and
  - ✓ a twin boundary

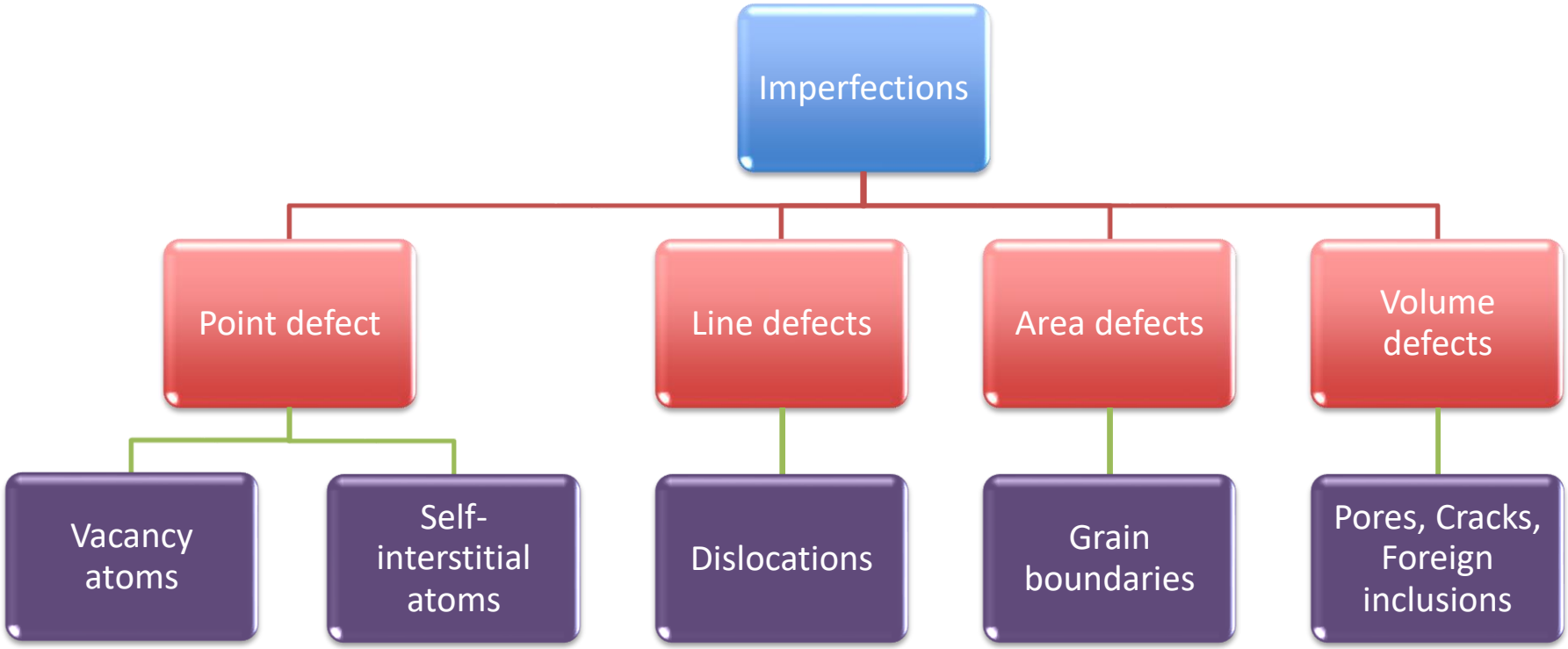
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Introduction

- **Perfect order does not exist** in crystalline materials at the atomic level.
- Crystalline materials contain large numbers of defects or imperfections.
- Many of the properties of materials are profoundly sensitive to deviations from crystalline perfection.
- Crystalline defect refers to a lattice irregularity having one or more of its dimensions on the order of an atomic diameter.
- Classification of crystalline imperfections is frequently made according to ***geometry or dimensionality*** of the defect.

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Types of Imperfections



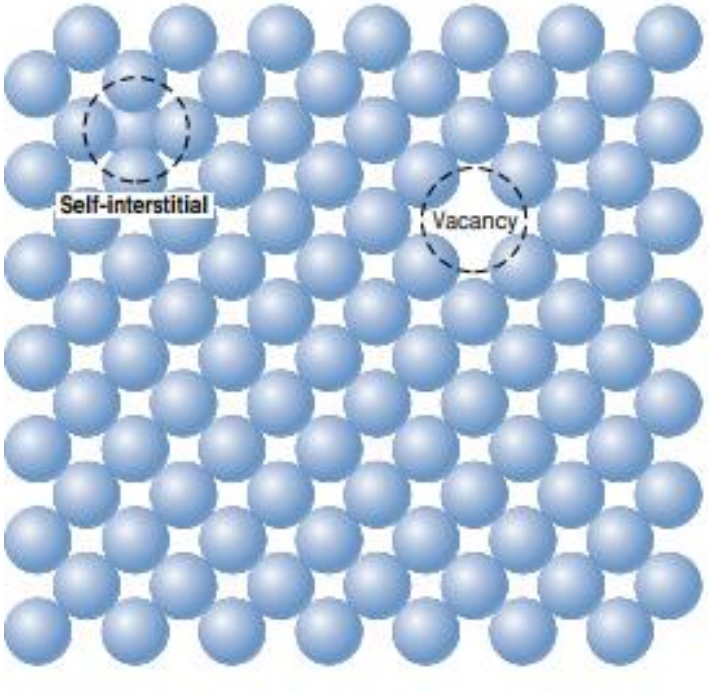
Type of Imperfections

- Point defects (0 Dimensional)
  - ✓ vacancies
  - ✓ interstitials
  - ✓ impurities, weight and atomic composition
- Dislocations (1 Dimensional)
  - ✓ edge
  - ✓ screw
- Grain boundaries (2 Dimensional)
  - ✓ tilt
  - ✓ twist
- Bulk or Volume defects (3 Dimensional)
  - ✓ Atomic vibrations

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Point Defects - Vacancies

- Vacancy is the absence of an atom from its normal location in a perfect crystal structure.
- All crystalline solids contain vacancies.
- The presence of vacancies increases the entropy of the crystal.



Temperature dependence of the equilibrium number of Vacancies

- The equilibrium number of vacancies for a given quantity of material depends on and increases with temperature according to

$$N_v = N \cdot exp\left(-\frac{Q_v}{kT}\right)$$

Where:

N is the total number of atomic sites

Q_v is the energy required for the formation of a vacancy,

T is the absolute temperature in kelvins

k is the gas or Boltzmann's constant

(k is 1.38 1023 J/atom·K or 8.62 105 eV/atom·K, depending on the units of Q_v)

Example

Calculate the equilibrium number of vacancies per cubic meter for copper at 1000℃. The energy for vacancy formation is 0.9 eV/atom; the atomic weight and density (at 1000 ℃) for copper are 63.5 g/mol and 8.4 g/cm^3, respectively.

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Solution

First of all, we need to determine the value of N, the number of atomic sites per cubic meter for copper, from its atomic weight A_{cu}, its density ρ, and Avogadro's number N_A, according to

$$N = \frac{N_A \cdot \rho}{A_{cu}}$$

$$= \frac{(6.022 \times 10^{23}\ atoms/mol)(8.4\ g/cm^3)(10^6\ cm^3/m^3)}{63.5\ g/mol}$$

$$N = 8.0 \times 10^{28}\ atoms/m^3$$

Thus, the equilibrium number of vacancies at 1000 °C (1273 K) is equal to

$$n = N \cdot exp\left(\frac{-Q}{kT}\right)$$

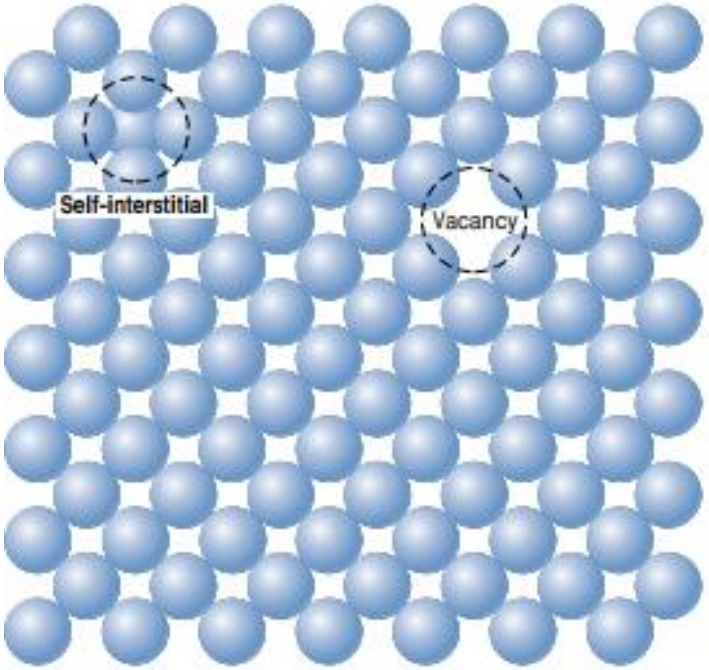
$$= 8.0 \times 10^{28}\ atoms/m^3 \cdot exp\left(\frac{-(0.9\ eV/atom)}{(8.62 \times 10^{-5}\ eV/atom \cdot K)(1273\ K)}\right)$$

$$n = 2.2 \times 10^{25}\ vacancies/m^3$$

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Self-interstitial

- A self-interstitial is an atom from the crystal that is crowded into an interstitial site, a small void space that under ordinary circumstances is not occupied.



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Impurities In Solids

- A pure metal consisting of only one type of atom just isn't possible; impurity or foreign atoms will always be present
- Even with relatively sophisticated techniques, it is difficult to refine metals to a purity in excess of 99.9999%.
- Most familiar metals are not highly pure; rather, they are **alloys,** in which impurity atoms have been added intentionally to impart specific characteristics to the material.
- Ordinarily alloying is used in metals to improve mechanical strength and corrosion resistance.

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Impurities In Solids

- For example, sterling silver is a 92.5% silver/7.5% copper alloy. In normal ambient environments, pure silver is highly corrosion resistant, but also very soft. Alloying with copper significantly enhances the mechanical strength without depreciating the corrosion resistance appreciably.
- The addition of impurity atoms to a metal will result in the formation of a **solid solution** and/or a new second phase, depending on the kinds of impurity, their concentrations, and the temperature of the alloy.

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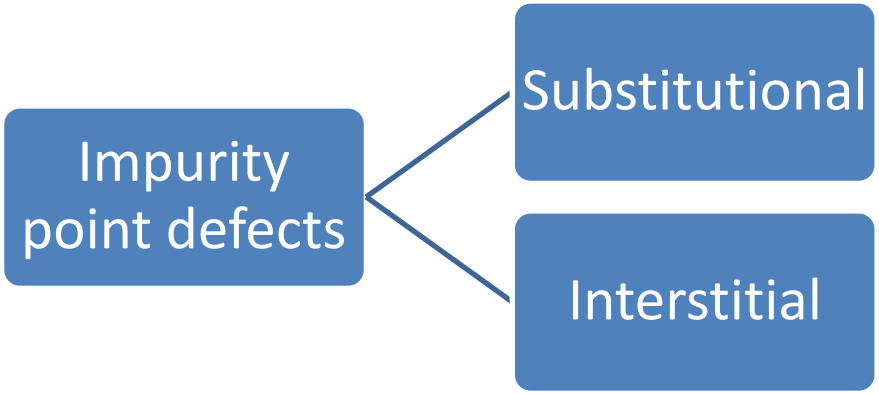
Impurities In Solids

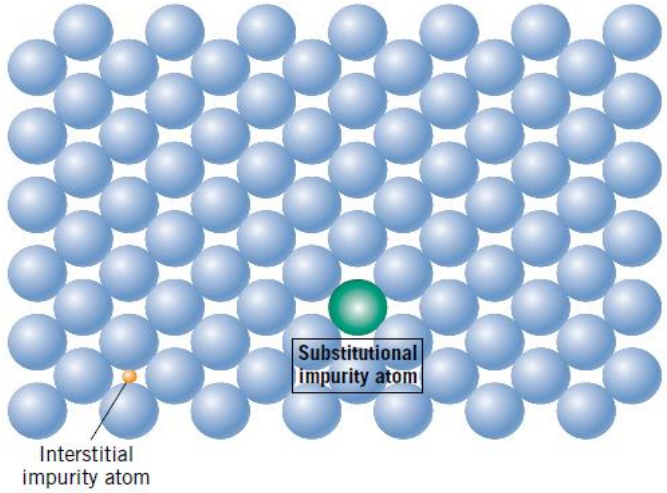
- With regard to alloys, **solute** and **solvent** are terms that are commonly employed.
- ***Solvent*** represents the element or compound that is present in the greatest amount; on occasion, solvent atoms are also called ***host atoms***.
- ***Solute*** is used to denote an element or compound present in a minor concentration.
- A **solid solution** forms when, as the solute atoms are added to the host material, the crystal structure is maintained and no new structures are formed.
- A solid solution is also compositionally homogeneous; the impurity atoms are randomly and uniformly dispersed within the solid

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Impurities in solid solution

- Impurity point defects are found in solid solutions, of which there are two types: **substitutional** and **interstitial.**
- **Substitutional solid solutions**: solute or impurity atoms replace or substitute for the host atoms.
- **Interstitial solid solutions**: impurity atoms fill the voids or interstices among the host atoms. The maximum allowable concentration of interstitial impurity atoms is low (less than 10%).





Conditions for Substitutional Solid Solution

1. **Atomic size factor**. Appreciable quantities of a solute may be accommodated in this type of solid solution only when the difference in atomic radii between the two atom types is less than about ±15%. Otherwise the solute atoms will create substantial lattice distortions and a new phase will form.
2. **Crystal structure**. For appreciable solid solubility the crystal structures for metals of both atom types must be the same.
3. **Electronegativity**. The more electropositive one element and the more electronegative the other, the greater the likelihood that they will form an intermetallic compound instead of a substitutional solid solution.
4. **Valences**. Other factors being equal, a metal will have more of a tendency to dissolve another metal of higher valence than one of a lower valence.

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Specification of Composition

- The **composition** (or *concentration*)[3] of an alloy is mostly expressed in terms of its constituent elements.
- The Composition is expressed as **weight (or mass) percent** or **atom percent**.
- **In weight percent:**

$$C_1 = \frac{m_1}{m_1 + m_2} \text{ x } 100$$

Where: C_1 is concentration of element 1

m_1 and m_2 represent the weight (or mass) of elements 1 and 2, respectively.

Specification Of Composition

- **In atom percent**:

$$C_1' = \frac{n_{m1}}{n_{m1} + n_{m2}} \times 100$$

Where: $\acute{C}_1$ is concentration of element 1

n_{m1} and n_{m2} represent number of moles in some specified mass of element 1

- Where $n_{m1} = \frac{\acute{m}_1}{A_1}$ and $n_{m2} = \frac{\acute{m}_2}{A_2}$
- Also $\acute{m}_1$ and A_1 denote the mass (in grams) and atomic weight, respectively, for element 1.
- $\acute{m}_2$ and A_2 denote the mass (in grams) and atomic weight, respectively, for element 2.

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Composition conversion

- **Weight % to Atomic %**

$$C_1^{at} = \frac{C_1^{wt} A_2}{C_1^{wt} A_2 + C_2^{wt} A_1} \times 100$$

$$C_2^{at} = \frac{C_2^{wt} A_1}{C_1^{wt} A_2 + C_2^{wt} A_1} \times 100$$

- **Atomic % to Weight %:**

$$C_1^{wt} = \frac{C_1^{at} A_1}{C_1^{at} A_1 + C_2^{at} A_2} \times 100$$

$$C_2^{wt} = \frac{C_2^{at} A_2}{C_1^{at} A_1 + C_2^{at} A_2} \times 100$$

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Composition conversion

- **From Weight % to mass per unit volume (g/cm^3):**

$$C_1 = \frac{C_1^{wt}}{\frac{C_1^{wt}}{\rho_1} + \frac{C_2^{wt}}{\rho_2}} \qquad C_2 = \frac{C_2^{wt}}{\frac{C_1^{wt}}{\rho_1} + \frac{C_2^{wt}}{\rho_2}}$$

C_1 and C_2 are concentrations of the first and second components in g/cm^3

- **Average density of a binary alloy**

$$\rho_{ave} = \frac{100}{\frac{C_1^{wt}}{\rho_1} + \frac{C_2^{wt}}{\rho_2}}$$

- **Average atomic weight of a binary alloy**

$$A_{ave} = \frac{C_1^{at} A_1 + C_2^{at} A_2}{100}$$

Example

Determine the composition, in atom percent, of an alloy that consists of 97 wt% aluminum and 3 wt% copper. Take atomic weight of copper and aluminum as 63.55 g/mol and 26.98 g/mol respectively.

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Solution

If we denote the respective weight percent compositions as $C_{Al} = 97$ and $C_{Cu} = 3$, then substituting into the atom percent equation,

$$\acute{C}_{Al} = \frac{C_{Al}A_{Cu}}{C_{Al}A_{Cu} + C_{Cu}A_{Al}} \times 100$$

$$= \frac{97 \times 63.55}{(97 \times 63.55) + (3 \times 26.98)} \times 100$$

$$= 98.7 \text{ at\%}$$

Solution cont.

$$\acute{C}_{Cu} = \frac{C_{Cu}A_{Al}}{C_{Cu}A_{Al} + C_{Al}A_{Cu}} \times 100$$

$$= \frac{3 \times 26.98}{(3 \times 26.98) + (97 \times 63.55)} \times 100$$

$$= \ 1.30 \text{ at } \%$$

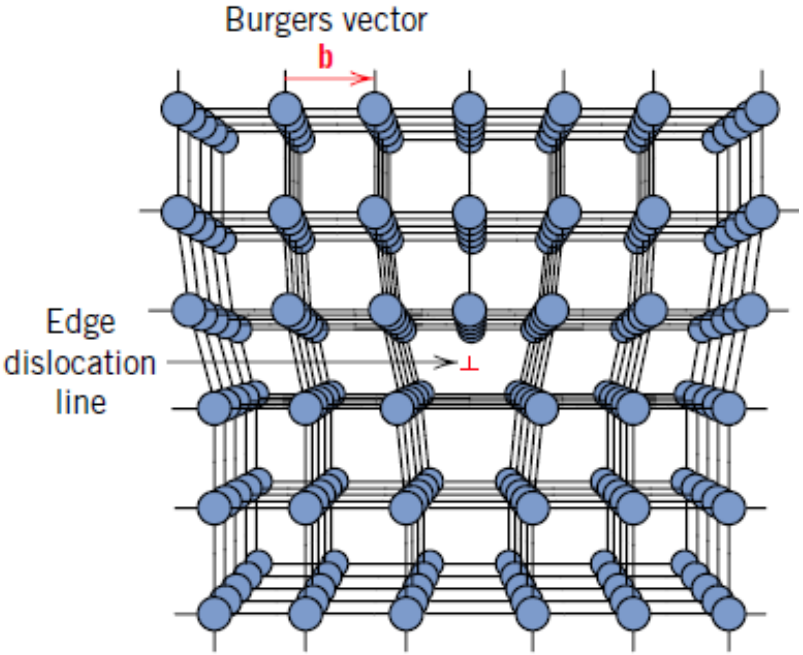
Line Defects - Burgers vector (b)

- The magnitude and direction of the lattice distortion associated with a dislocation is expressed in terms of a **Burgers vector,** denoted by a **b.**
- Furthermore, the nature of a dislocation (i.e., edge, screw, or mixed) is defined by the relative orientations of dislocation line and Burgers vector.
- For an edge, they are perpendicular whereas for a screw, they are parallel; they are neither perpendicular nor parallel for a mixed dislocation.

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Linear Defects - Dislocations

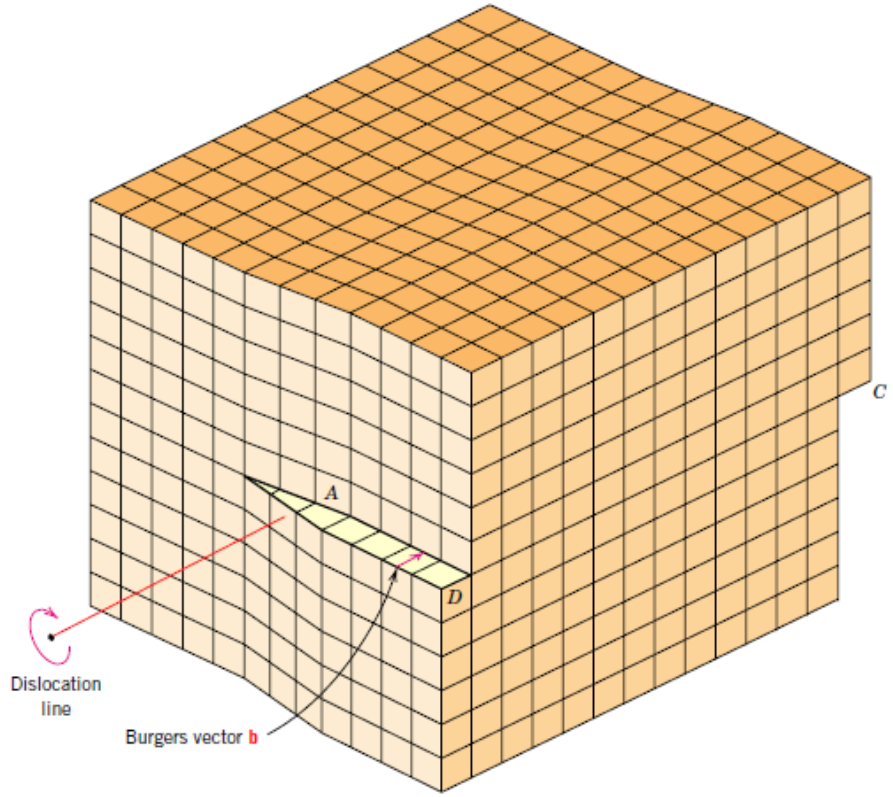
- A **dislocation** is a linear or one-dimensional defect around which some of the atoms are misaligned.
- Can be **edge dislocation** or **screw dislocation.**
- **Edge dislocation:** Extra half-plane of atoms inserted in a crystal structure. Burgers vector **b** is perpendicular (⊥) to dislocation line.



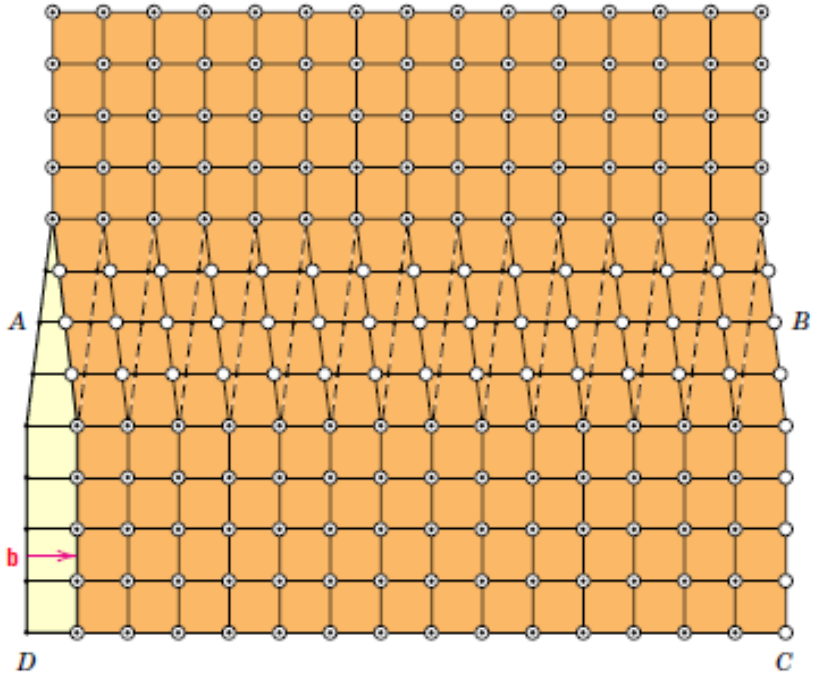
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Screw Dislocations

- **Screw dislocation**: Spiral planar ramp resulting from shear deformation. Burgers vector **b** is parallel (||) to dislocation line.



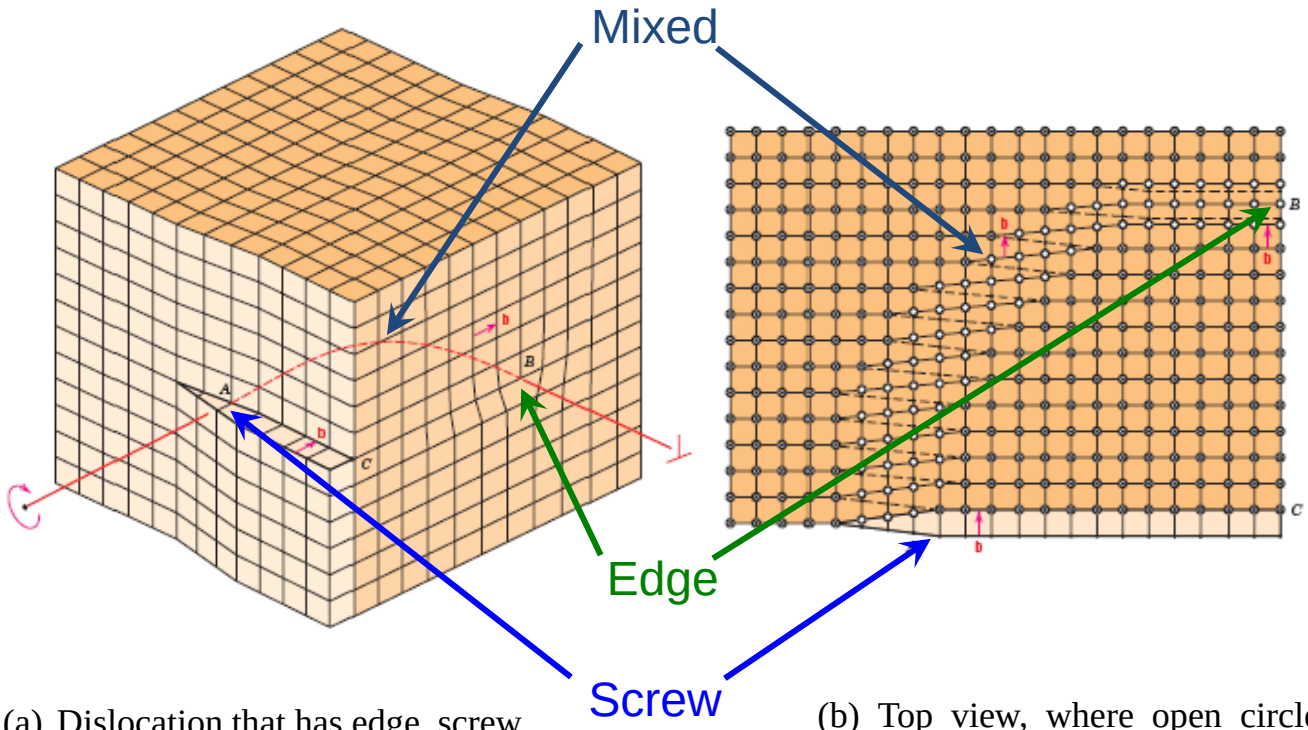
(a) Screw dislocation within a crystal



(b) The screw dislocation in (a) as viewed from above

Mixed Dislocations

- **Mixed dislocation**: Most dislocations found in crystalline materials are probably neither pure edge nor pure screw, but exhibit components of both types; these are termed **mixed dislocations.**



(a) Dislocation that has edge, screw, and mixed character

(b) Top view, where open circles denote atom positions above the slip plane, and solid circles, atom positions below slip plane

Area Defects - Interfacial Defects

- Interfacial defects are boundaries that have two dimensions and normally separate regions of the materials that have different crystal structures and/or crystallographic orientations.
- These imperfections include **external surfaces, grain boundaries, phase boundaries, twin boundaries, and stacking faults.**

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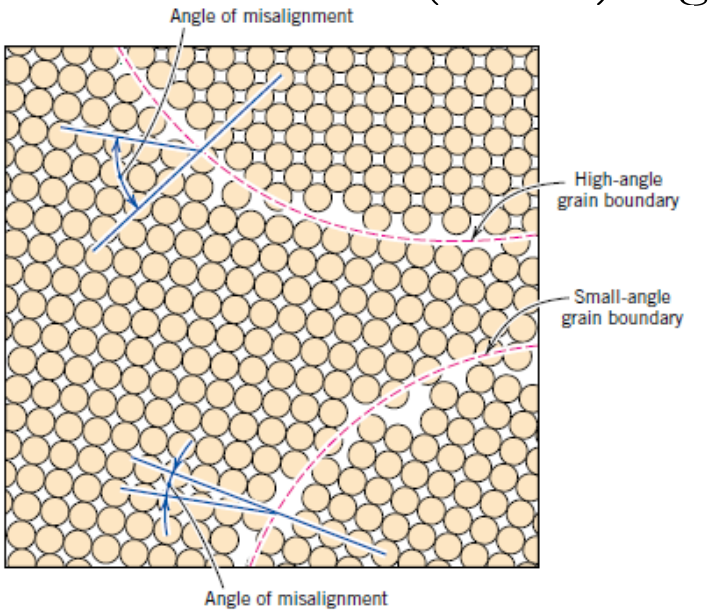
External Surfaces

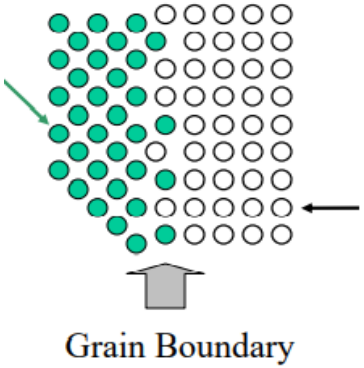
- One of the most obvious boundaries is the external surface, along which the crystal structure terminates.
- Surface atoms are not bonded to the maximum number of nearest neighbours, and are therefore in a higher energy state than the atoms at interior positions.
- The bonds of these surface atoms that are not satisfied give rise to a surface energy, expressed in units of energy per unit area (J/m^2 or erg/cm^2).
- To reduce this energy, materials tend to minimize, if at all possible, the total surface area. For example, liquids assume a shape having a minimum area - the droplets become spherical. Of course, this is not possible with solids, which are mechanically rigid.

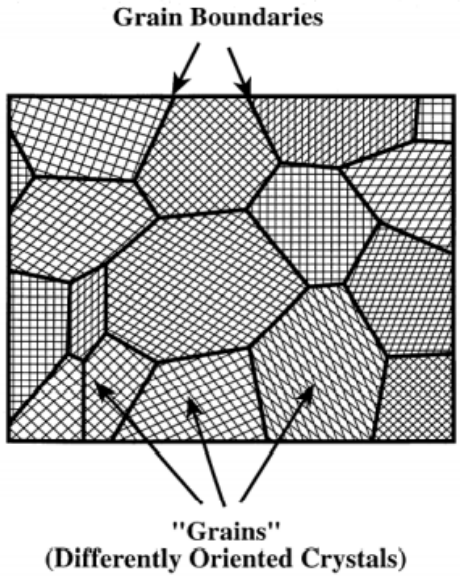
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Grain boundaries

- The boundary separating two small grains or crystals having different crystallographic orientations in polycrystalline materials.
- Various degrees of crystallographic misalignment between adjacent grains are possible.
- When this orientation mismatch is slight, on the order of a few degrees, then the term *small-* (or *low-*) *angle grain boundary* is used.







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Tilt boundaries

- Tilt boundary is a simple small-angle grain boundary that occurs when edge dislocations are aligned.
- Twist boundary occurs when the angle of mis-orientation is parallel to the boundary.

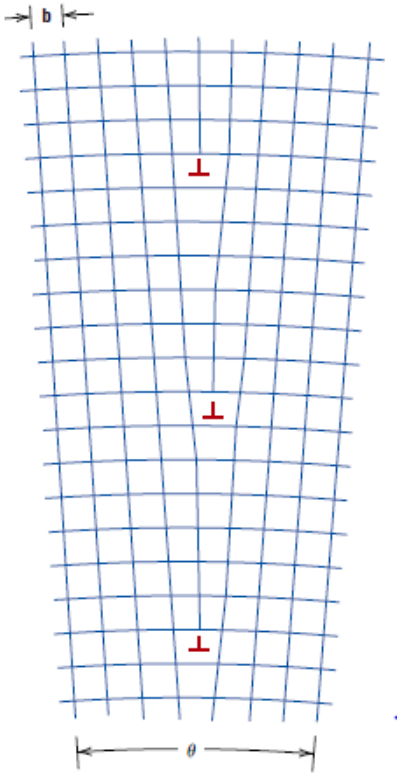


Fig. Tilt boundary having an angle of mis-orientation

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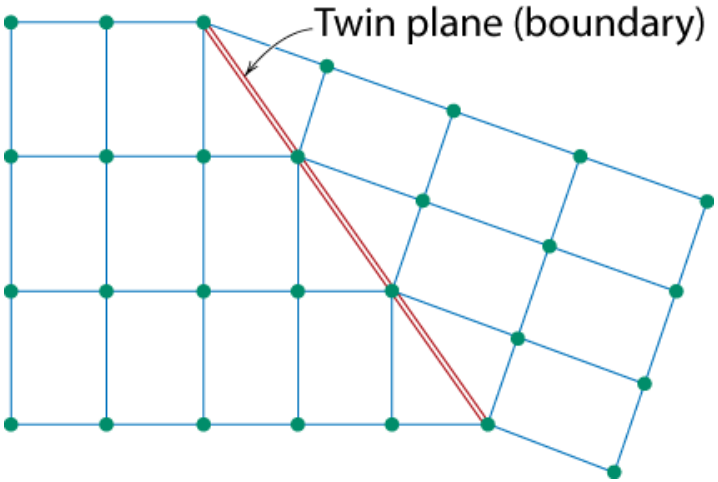
Phase Boundaries

- **Phase boundaries** exist in multiphase materials, wherein a different phase exists on each side of the boundary.
- Each of the constituent phases has its own distinctive physical and/or chemical characteristics. Phase boundaries play an important role in determining the mechanical characteristics of some multiphase metal alloys.

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Twin Boundaries

- A ***twin boundary*** is a special type of grain boundary across which there is a specific mirror lattice symmetry
- Atoms on one side of the boundary are located in mirror-image positions of the atoms on the other side.
- The region of material between these boundaries is appropriately termed a *twin*.
- Twins result from atomic displacements that are produced from applied mechanical shear forces (mechanical twins), and also during annealing heat treatments following deformation (annealing twins).



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Bulk or Volume Defects

- ***Pores***
- ***Cracks***
- ***Foreign inclusions***
- Presence of volume defects can greatly affect electrical, mechanical, thermal, and optical properties of the material.

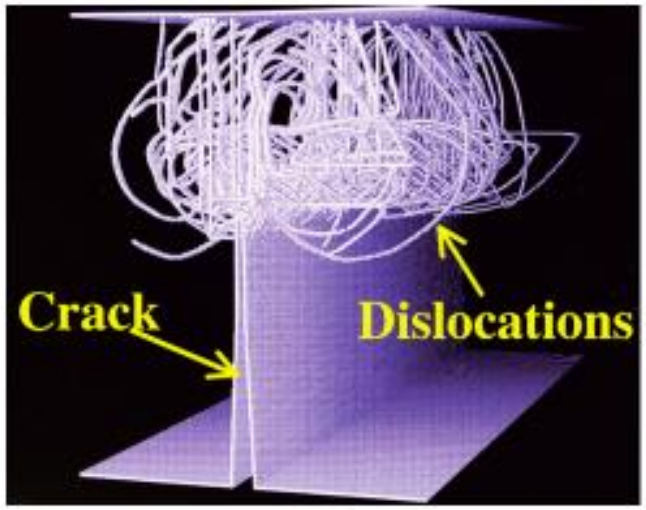


Fig. Atomistic simulation of crack propagation

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Atomic Vibrations

- Thermal energy (heat) causes atoms to vibrate
- Vibration amplitude increases with temperature
- Melting occurs when vibrations are sufficient to rupture bonds
- Vibrational frequency ~ 10^{13} Hz (10^{13} vibrations per second)
- Average atomic energy due to the thermal excitation is of order kBT

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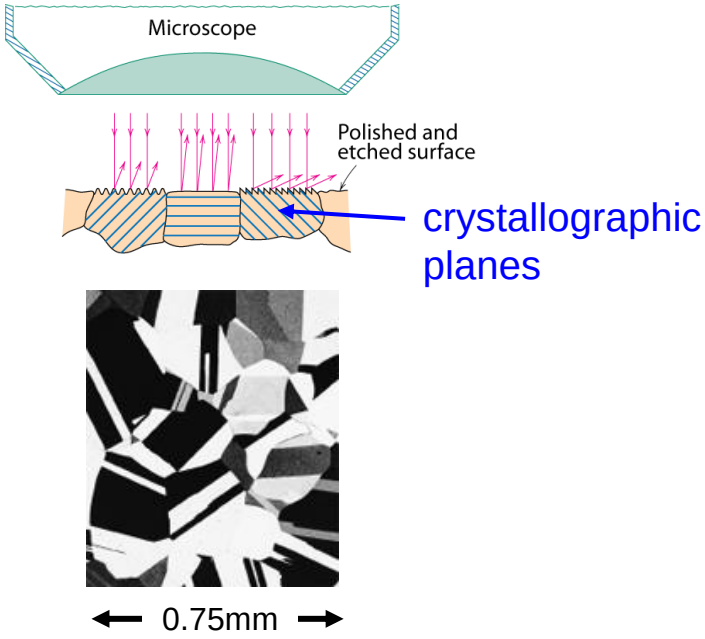
Microscopic Examination

- In most materials the grains are in microscopic dimensions, with diameters in the order of microns.
- Microstructure relates to grain size and shape.
- **Microscopic observations use optical, electron and scanning probe microscopes**.
- Photomicrographs are photographs recorded by microscopes.
- Microscopic examination is an extremely useful tool in the study and characterization of materials.
- The applications are:
  - ✓ To understand the relation between the properties and structure (and defects),
  - ✓ To predict the properties of materials,
  - ✓ To design alloys with new elements combination,
  - ✓ To determine whether or not a material has been correctly heat treated,
  - ✓ To determine the mode of mechanical fracture.

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Optical Microscopy

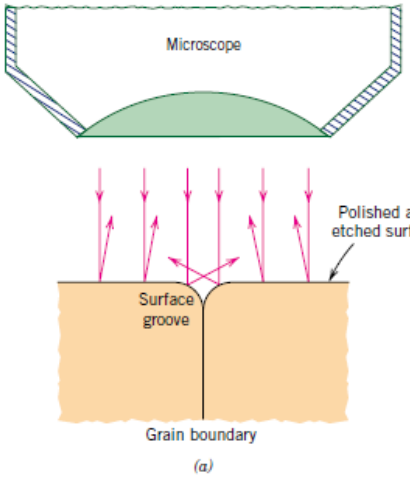
- Using optical and illumination systems.
- The upper limit of magnification is up to 2000 X
- Specimen surface must be ground and polished to a smooth and mirror-like finish.
- **Polishing** removes surface features (e.g., scratches)
- **Etching procedure**, especially for metal specimens changes reflectance, depending on crystal orientation

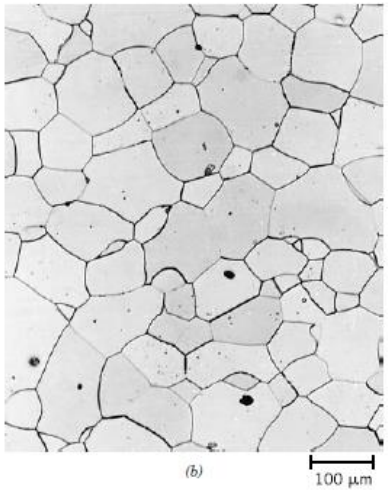


Micrograph of brass (a Cu-Zn alloy)

Electron Microscopy System

- Using accelerated electrons with l » 0.003 nm. The electron beam is focused and the image is formed with a magnetic lenses.
- ***Scanning Electron Microscope (SEM):*** The specimen surface is scanned with the electron beam and the reflected beam is collected then displayed on a CRT (Cathode Ray Tube).
- ***Transmission Electron Microscope (TEM):*** The electron beam passes through the specimen and the contrast of the internal microstructures are produced by differences in beam scattering produced between various elements of the microstructures.





Scanning Probe Microscope (SPM)

A tiny probe with a very sharp tip is brought into very close proximity (in the order of nm) to the specimen surface to generate a topographical map on atomic scale.

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Grain Size Determination

- Grain size often determines the properties of a polycrystalline materials. The grain size is specified in terms of grain volume, diameter or area.
- **Intercept Method:** Straight lines all the same length are drawn through several photomicrographs that show the grain structure. The grains intersected by each line segment are counted; the line length is then divided by an average number of grains intersected, taken over all the line segments. The average grain diameter is found by dividing this result by the linear magnification of the photomicrographs.

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Grain Size Determination

- ***Standard Comparison Charts Method:*** *ASTM Standard E112, "Standard Methods for Estimating the Average Grain Size for Metals".*
- Relationship between ASTM grain size number and number of grains per square inch (at 100) is given by

$$N = 2^{n-1}$$

- Where *n* is the grain size number and *N is* the average number of grains per square inch at a magnification of 100X

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Worked Example

Determine the ASTM grain size number of a metal specimen if 45 grains per square inch are measured at a magnification of 100.

Solution

a) In order to determine the ASTM grain size number (n) it is necessary to employ

$$N = 2^{n-1}$$

Taking logarithms of both sides of this expression leads to

$$logN = (n-1)log2$$

$$n = \frac{\log N}{log2} + 1$$

But N= 45, thus

$$n = \frac{log45}{log2} + 1 = 6.5$$

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Thank You